Algorithm 1 Training the Classifier and Reference-Free Generat

Input: clean labeled set D; target y*; trigger T; seen positions P_seen; loss weights lambda

Output: frozen suspicious classifier f_s and trained correction generator G_theta

D_p <- D
for each selected non-target sample (x,y) **do**
    sample attack position p and construct t <- T(x;p)
    D_p <- D_p union {(t,y*)}
end for
train f_s on D_p
verify clean accuracy, ASR lift, and conditional ASR; freeze f_s
initialize G_theta near the identity correction
for each generator epoch and clean minibatch (x,y) **do**
    sample p from P_seen; construct t <- T(x;p)
    Only t enters the generator. Clean x supplies the desired label and reconstruction target.
    (L_t,P_t) <- Spectrum(t); Z_t <- Evidence(L_t,P_t)
    Evidence describes amplitude, local anomaly, phase, and frequency location from t alone.
    (M,Delta) <- G_theta(Z_t); E <- M x Delta
    M decides where to intervene; Delta decides whether and how far amplitude moves.
    L_c <- max(L_t + E,0); x_c <- Reconstruct(L_c,P_t)
    Preserve the input phase and reconstruct the image using corrected log amplitude.
    x_id <- Correct(x;G_theta)
    L <- L_cls + 4L_img + L_spec + 2L_id + 0.02L_sp + 0.01L_sm
    Recover the true class while limiting image change, dense correction, and rough maps.
    update theta by minimizing L; keep f_s frozen
end for

Algorithm 2 Reference-Free Correction of One Inference Image

Input: one possibly attacked image u; frozen f_s; trained G_theta

Output: corrected image u_c; class prediction y_hat; correction evidence E

(L_u,P_u) <- Spectrum(u)
Z_u <- Evidence(L_u,P_u)

3: (M_u,Delta_u) <- G_theta(Z_u); E <- M_u x Delta_u

4: L_c <- max(L_u + E,0)

5: u_c <- Reconstruct(L_c,P_u)

6: y_hat <- arg max f_s(u_c)

7: **return** (u_c,y_hat,E)

Plain-language reading: training teaches a reusable correction from many controlled examples. At deployment, one new image supplies all spectral evidence; no clean twin or true label is available. The generator proposes one correction, reconstructs once, and the frozen classifier predicts once.
